

RELEBOM



SERENA LIN

CO-OP FALL



2024

START



About me!



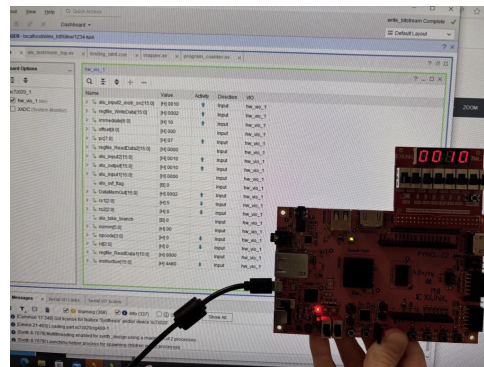
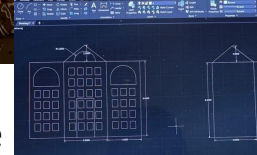
Hi I'm Serena Lin!

3rd Year ECE Student at
Northeastern

The question: EE or CE?

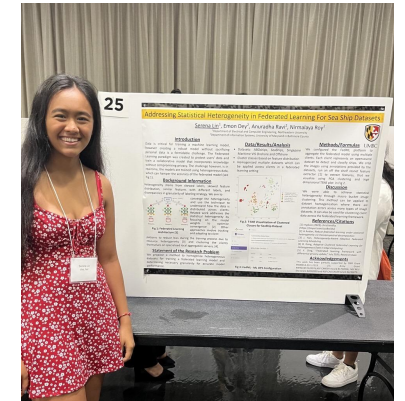
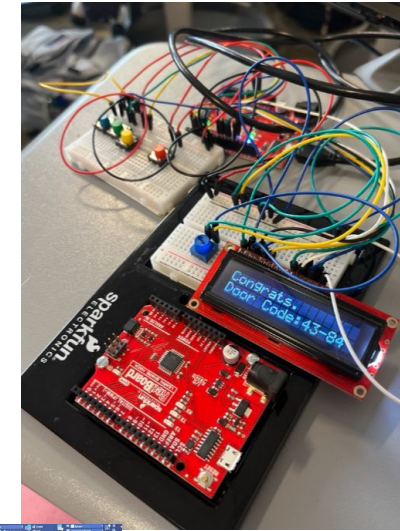


AutoCAD Designed
Gingerbread House

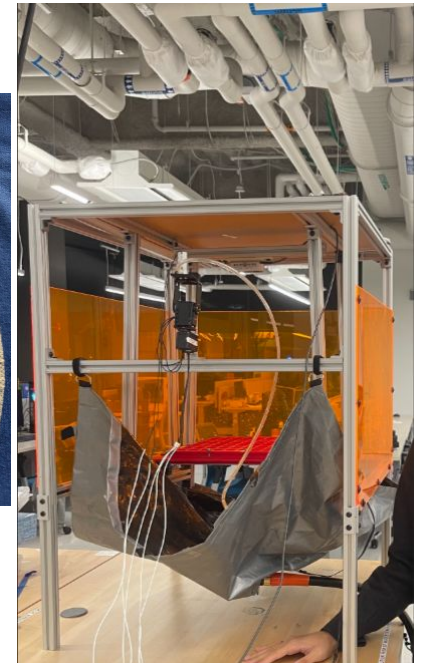


System Verilog
RISC5 Processor!

First ECE Project



First Poster
Presentation!



Autonomous Fluid
Projectile Robot



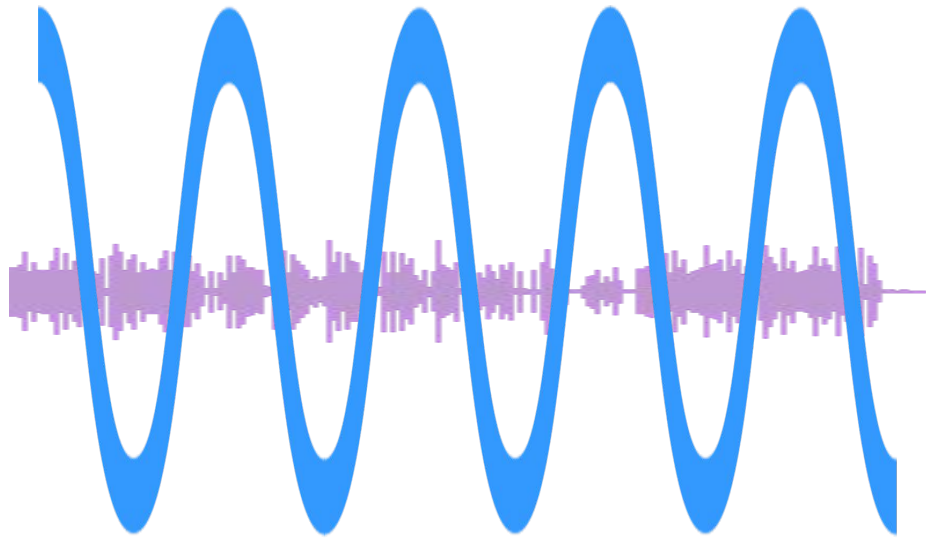
Recent
Embroidery!

Embedded
Robotic Arm!

DIFFERENTIAL MIC PREAMPLIFIER

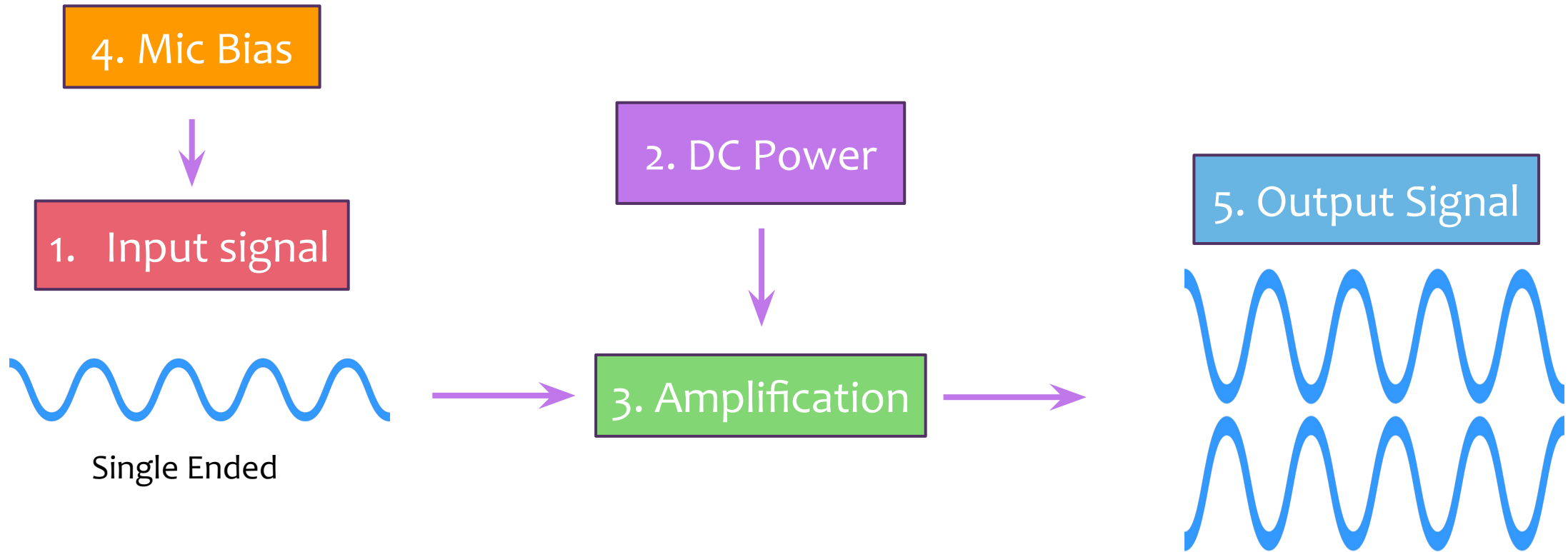
Mission: Design a new preamplifier to be compatible with differential microphones

What is the problem?



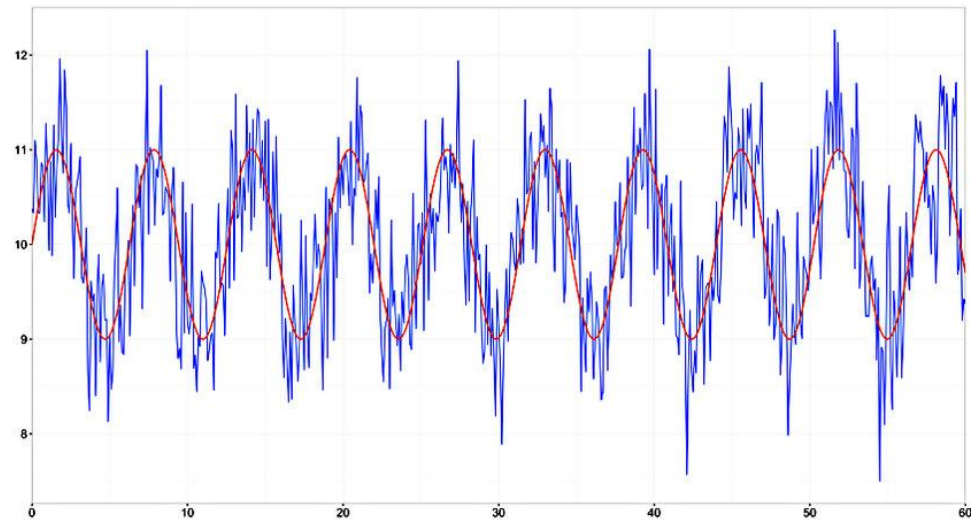
High noise floor at ground can cause excessive noise in small signals

Single Ended Microphone Pre-Amp Overview



Users & Purpose

- Ask: design a preamp compatible with differential mics
 - IM70A135 diff analog MEMS mic
 - ICS-40720 diff analog MEMS mic
- R&D customers
 - Andrew Munro
 - Richard Herr
- Easy to change bias voltage for mics
- ADCs do not support diff mics

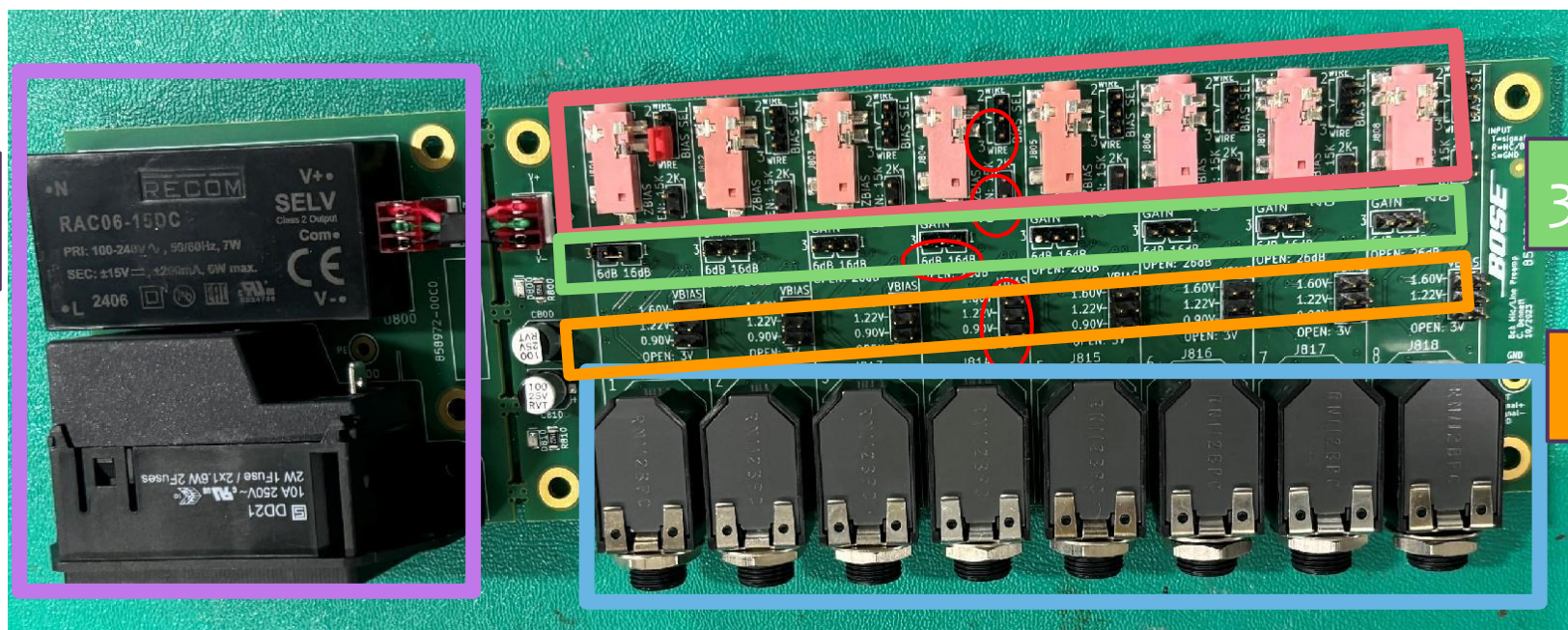


Single Ended Mic Preamp Overview

1. Input signal



2. DC Power



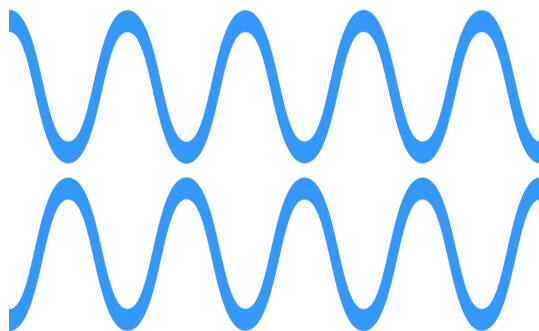
Gain : 6,16,26dB

3. Amplification

4. Mic Bias

Voltage Bias:
0.9,1,2,1.6,3V

5. Output Signal



History of the Pre-Amplifier

Revision A: single ended, 3 mic biases (1.2V, 1.6V, 3V)

Revision B: single ended, silkscreen modifications

Revision C: single ended, adds 4th mic bias (0.9V) + thicker PCB

Revision D: differential, differential microphones

Both revs C and D will continue to be used

C ☐ Single Ended Microphones

D ☐ Differential Microphones

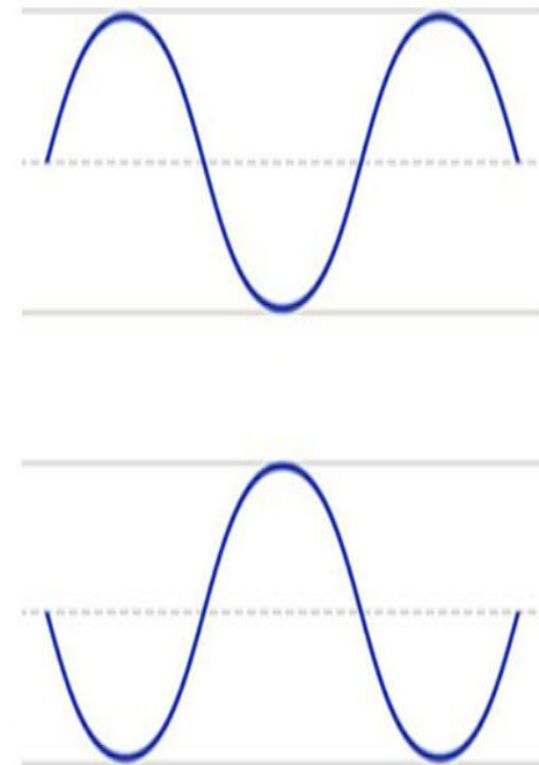
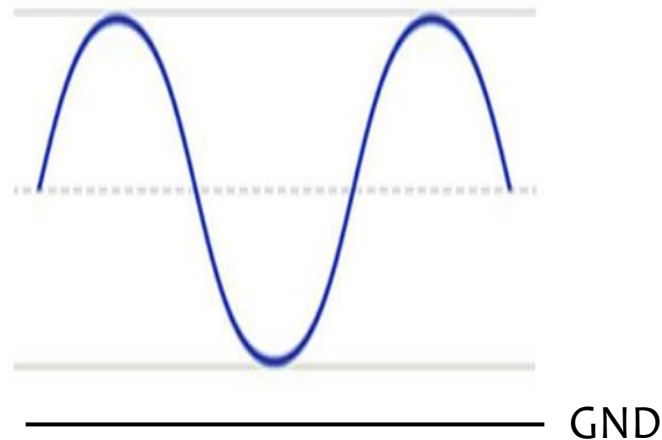


Single Ended Input ☐ Pseudo-Differential Output

input positive signal ☐

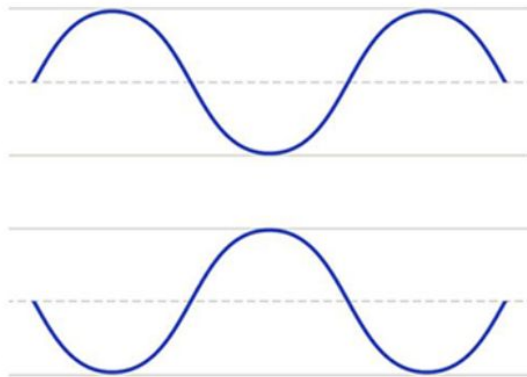
amplify signal ☐

duplicate + invert signal

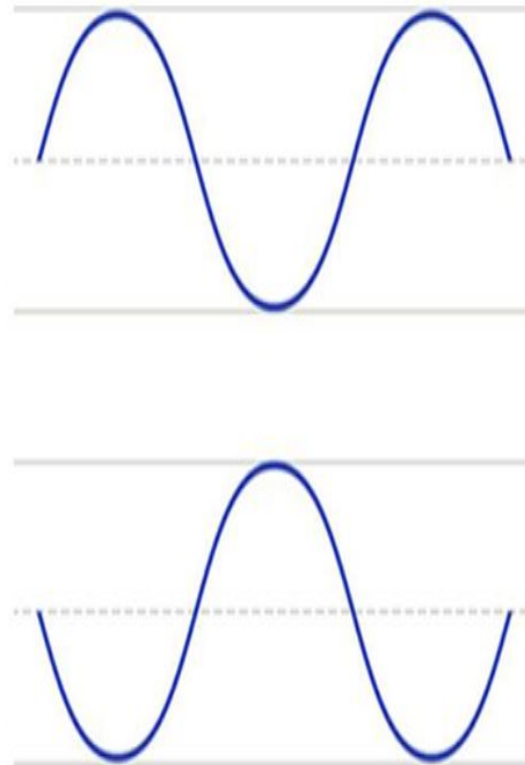


Differential Input □ Differential Output

input positive + negative signal □



amplify signal

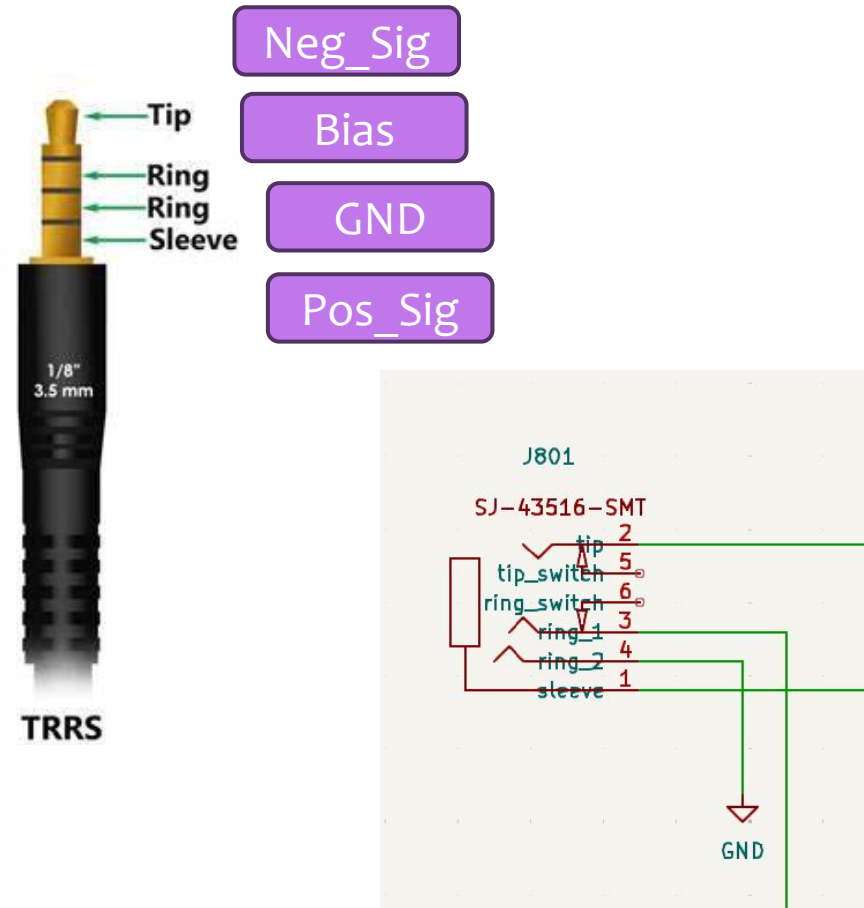
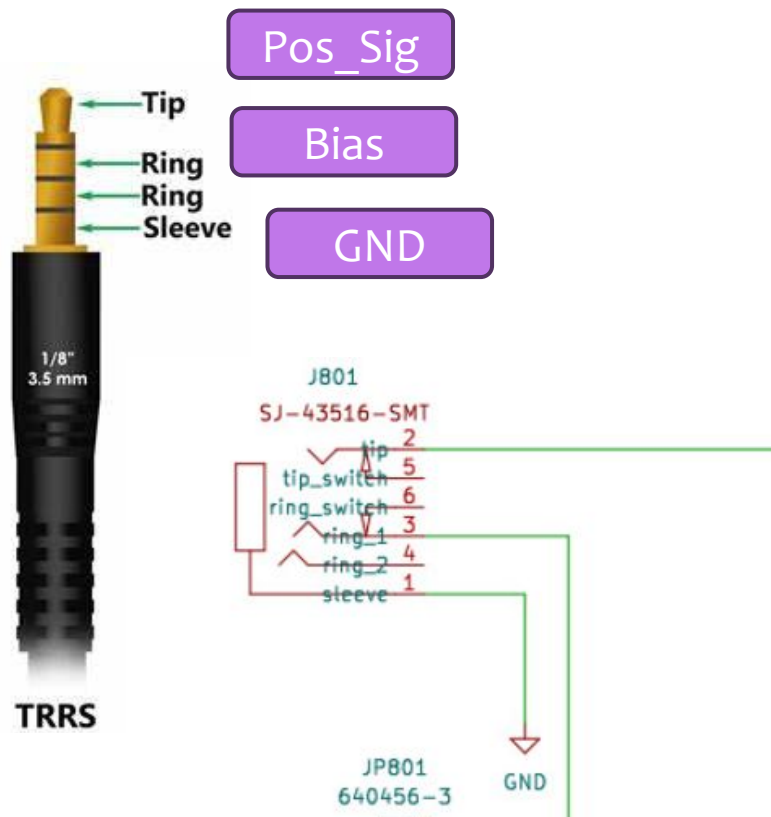


No comparison
to GND!

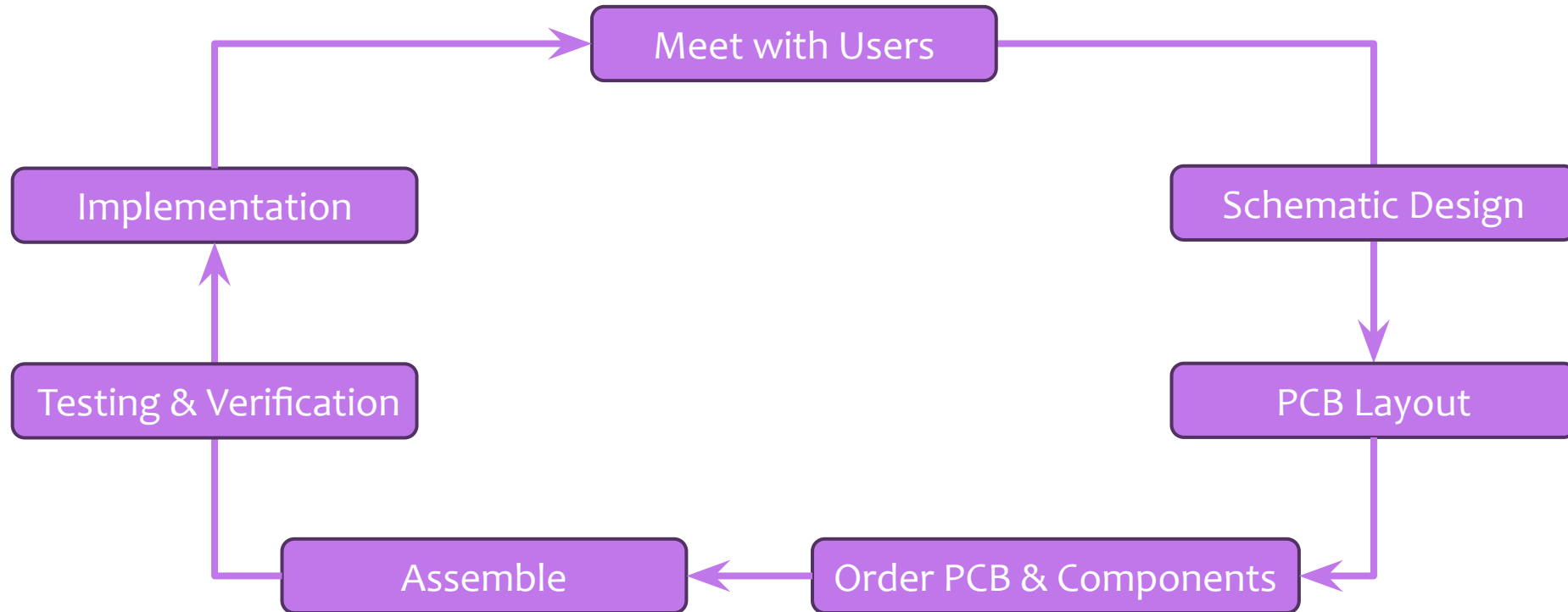
Single Ended

vs

Differential Mic

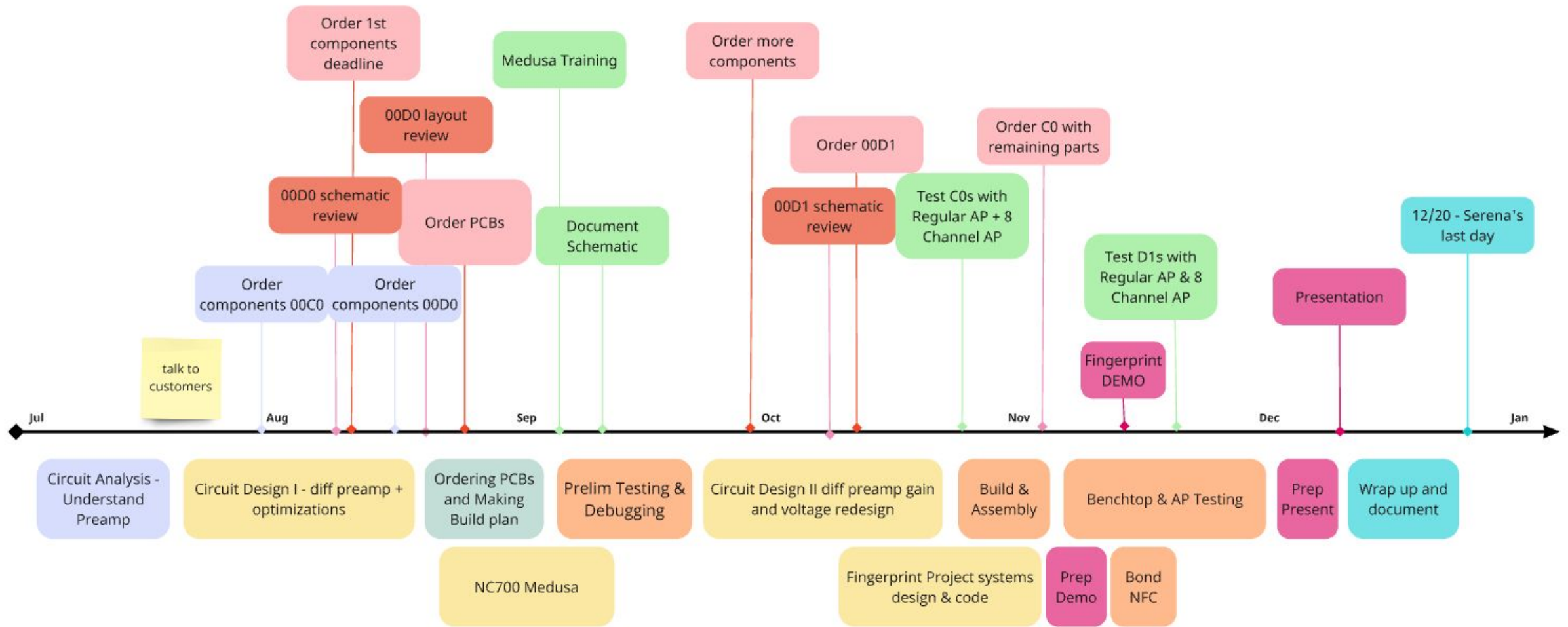


Design Process



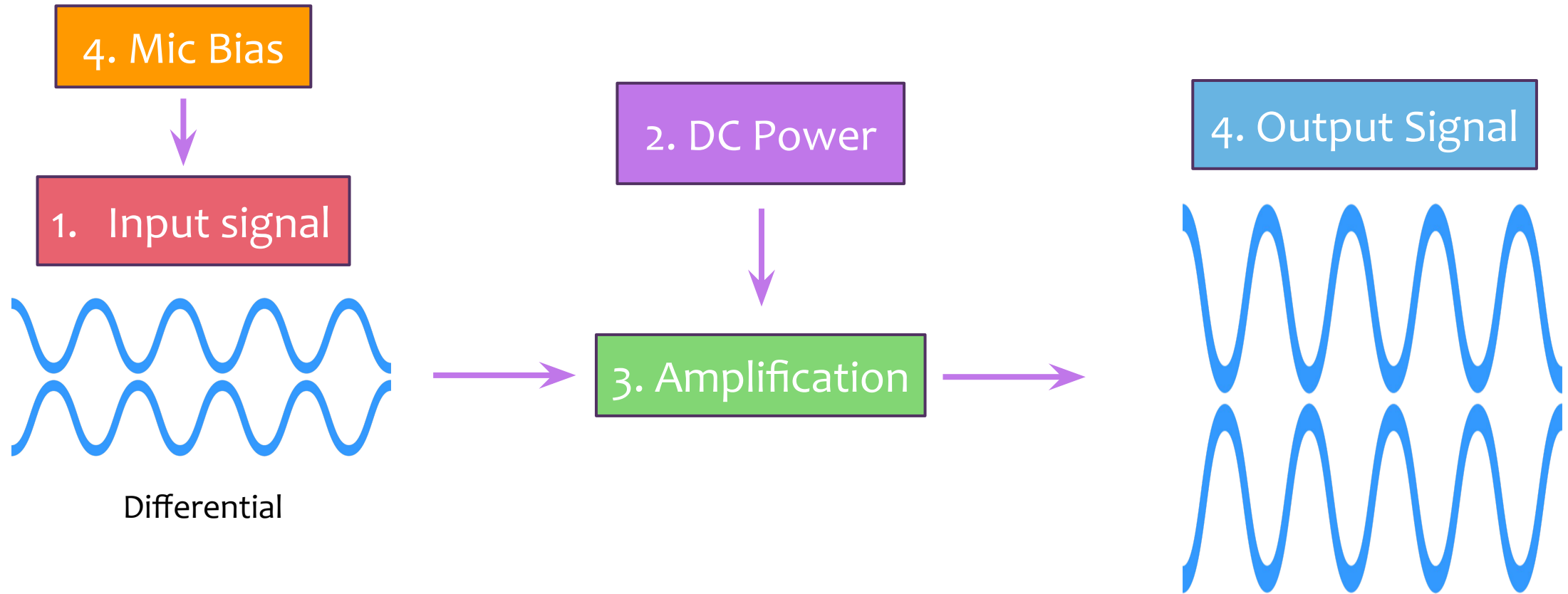
Using Plan Do Check & Act Model!

Managing the Project & Setting Deadlines



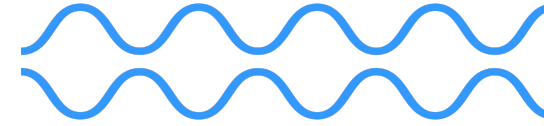
SCHEMATIC DESIGN

Differential Microphone Pre-Amp Overview

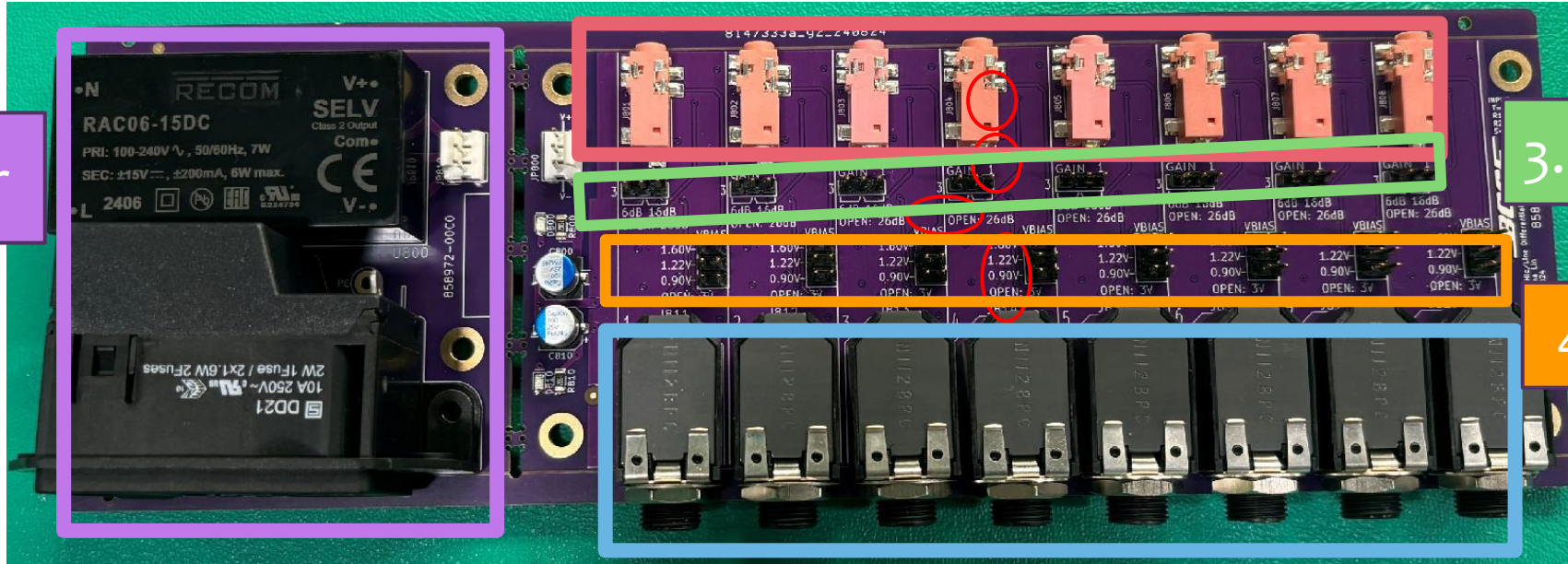


Differential Mic Preamp Overview

1. Input signal



2. DC Power



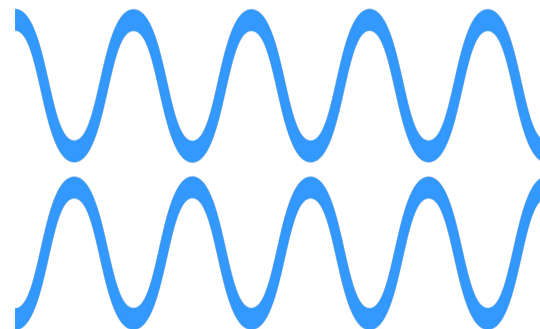
Gain : 6,16,26dB

3. Amplification

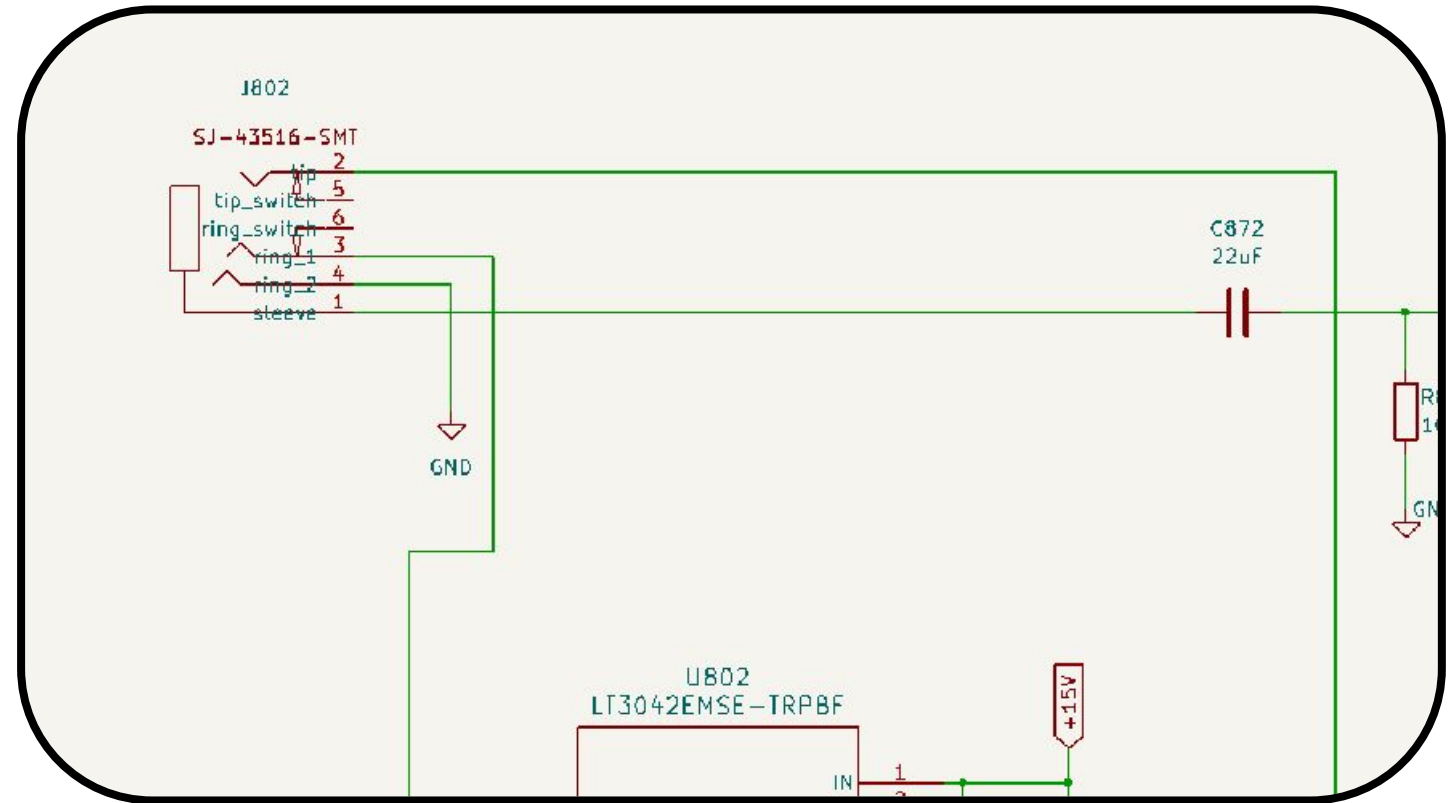
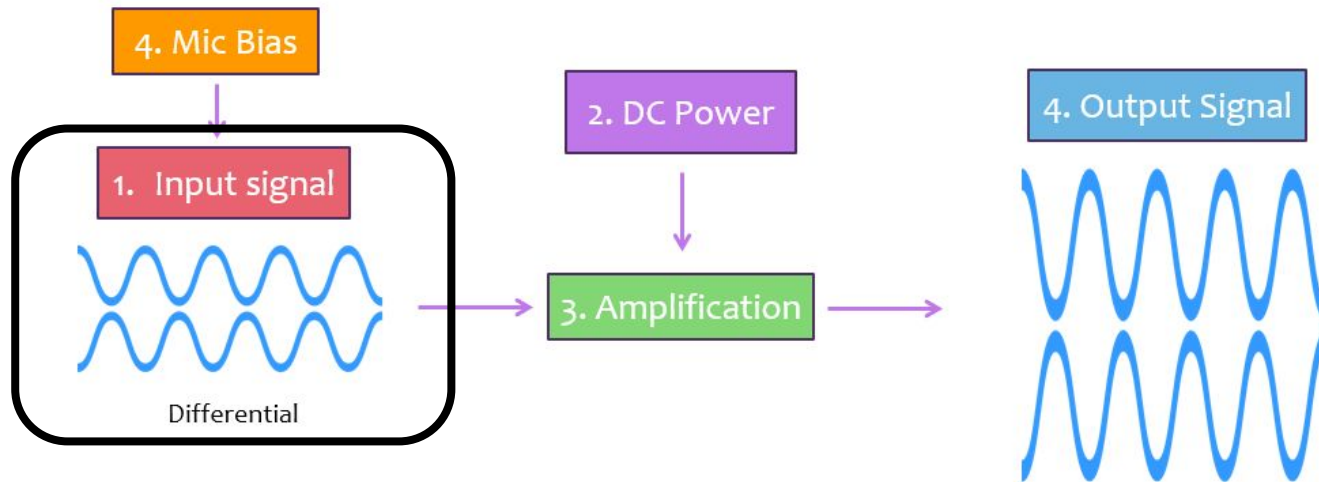
4. Mic Bias

Voltage Bias: 1.6V,
2,75V, 3V

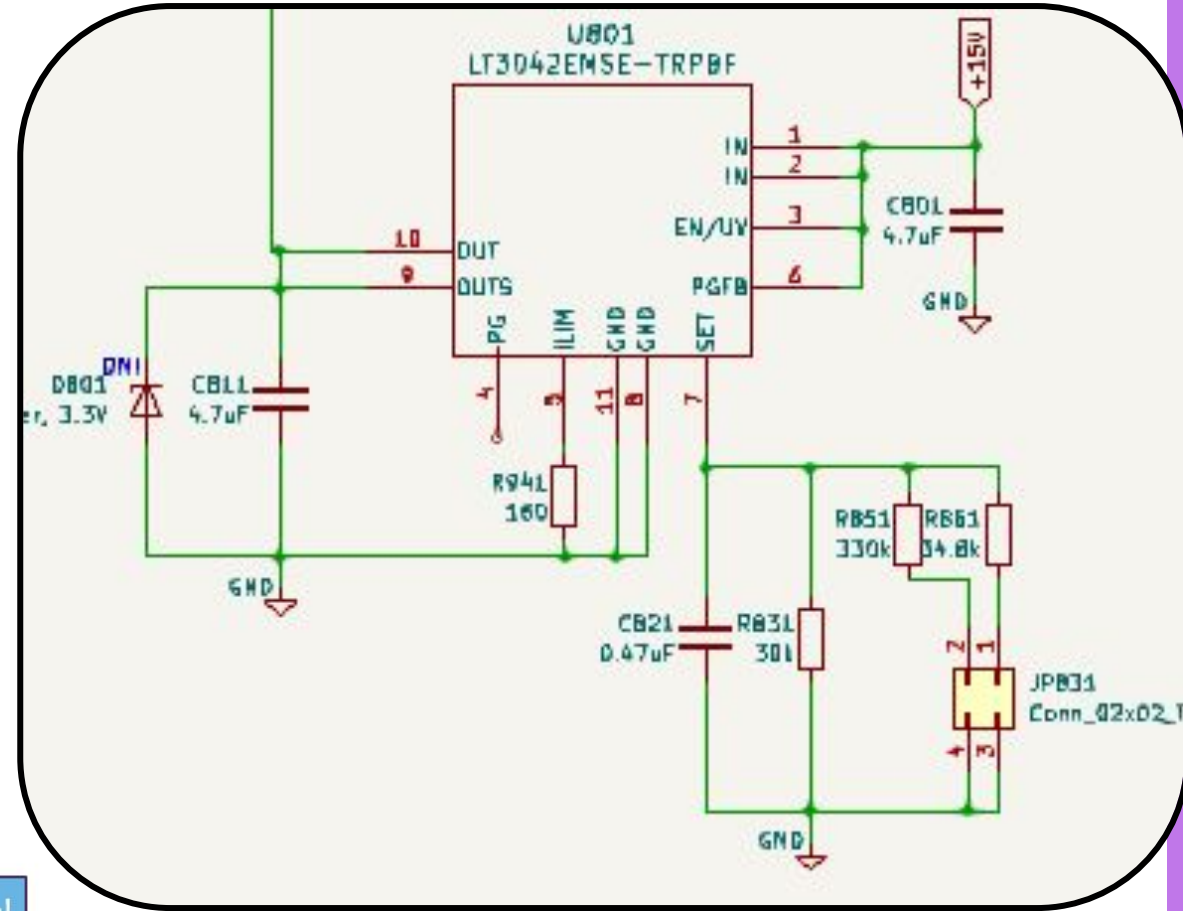
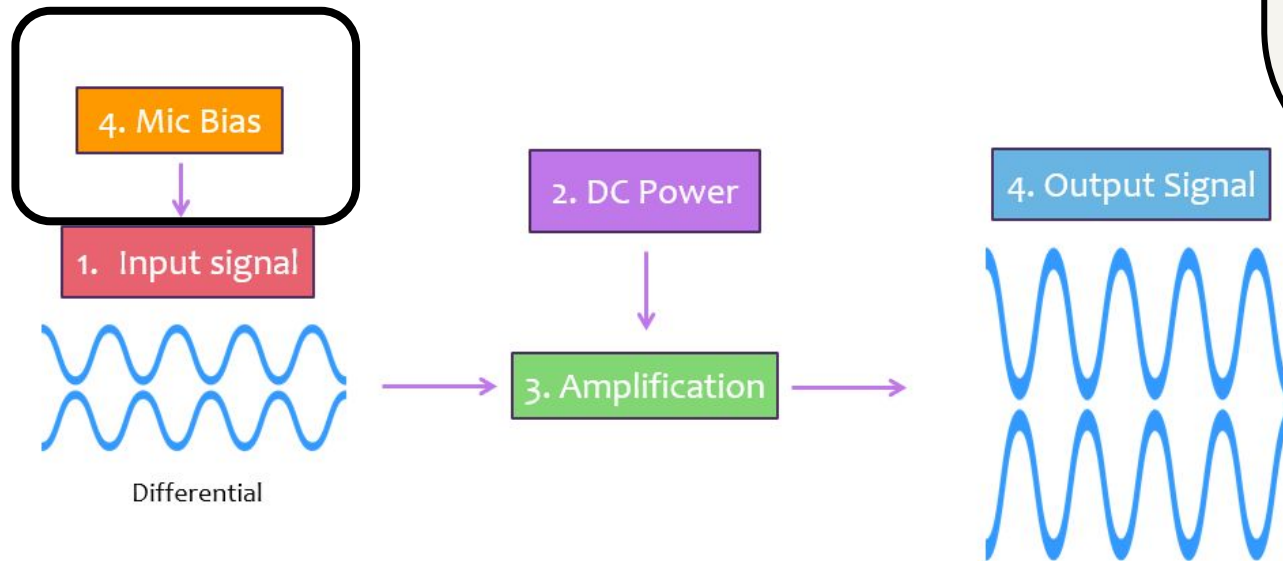
5. Output Signal



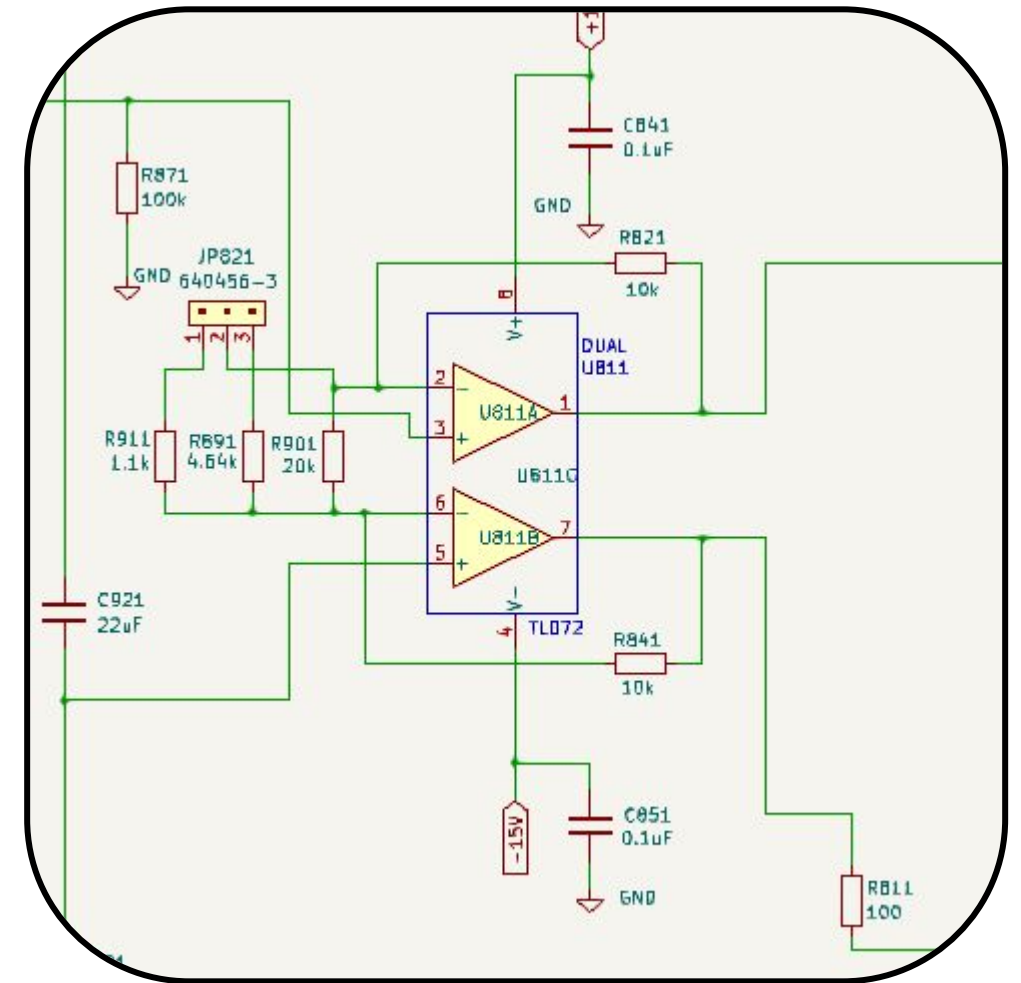
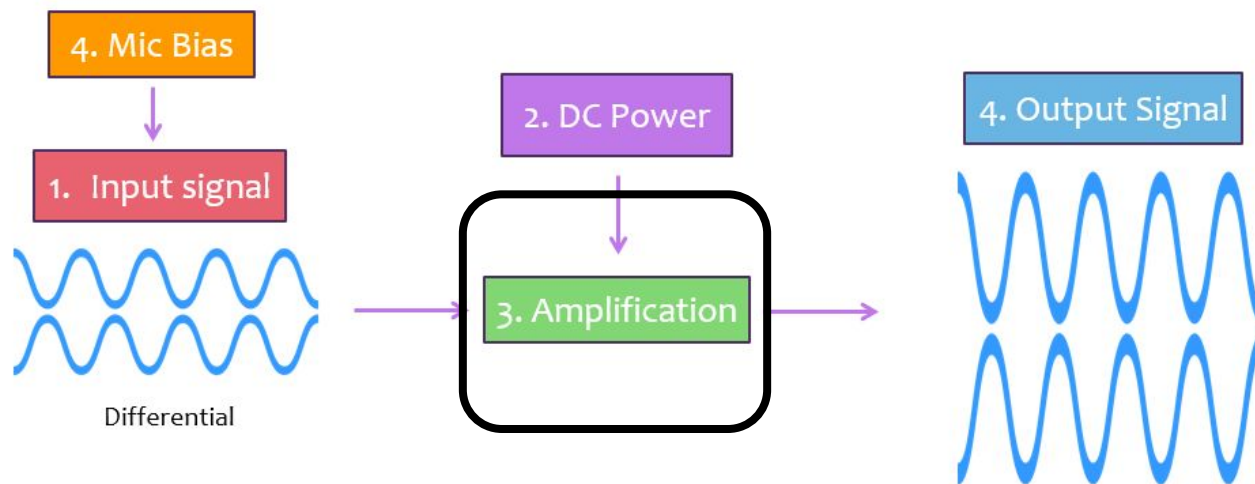
1. Input signal



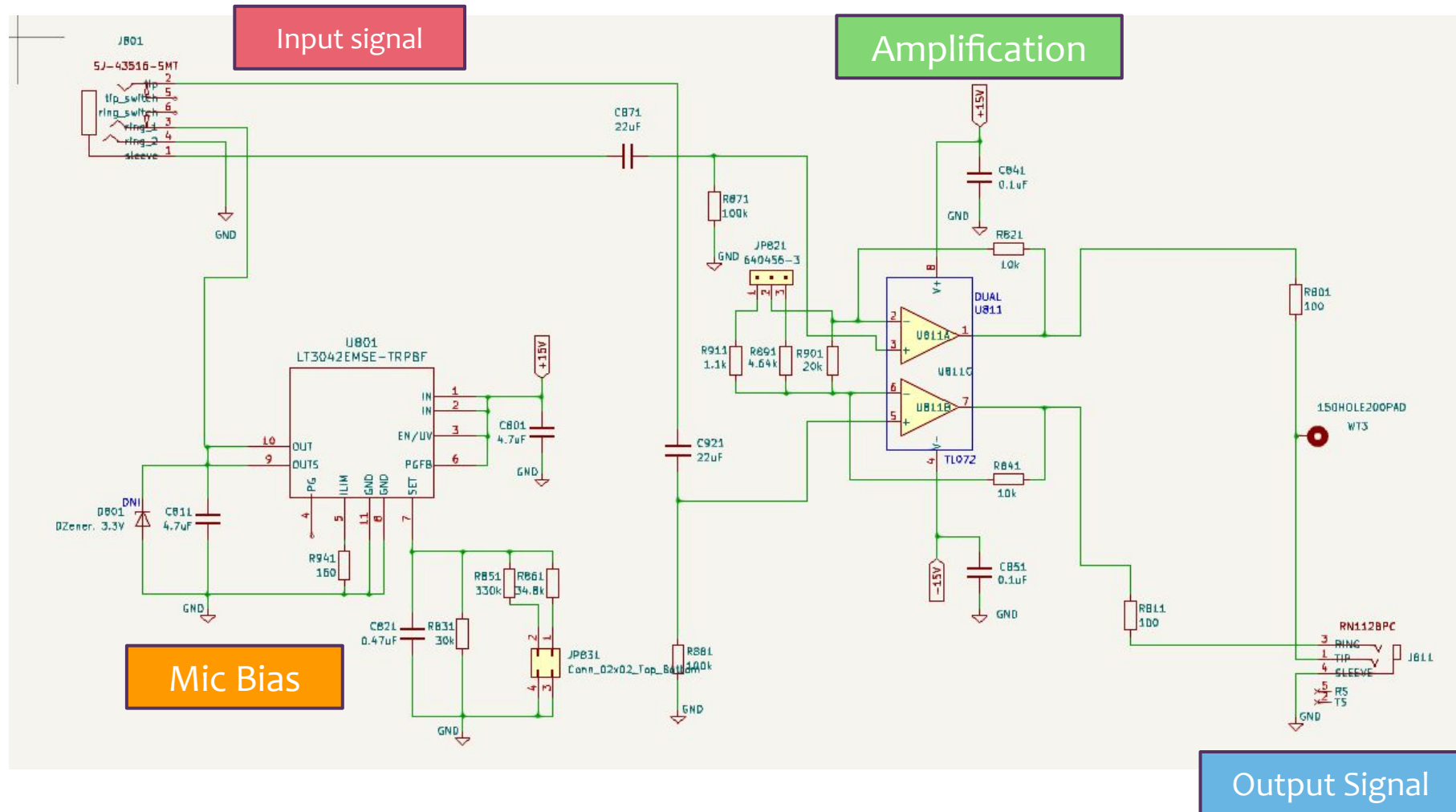
4. Mic Bias: Voltage Regulator



3. Amplification



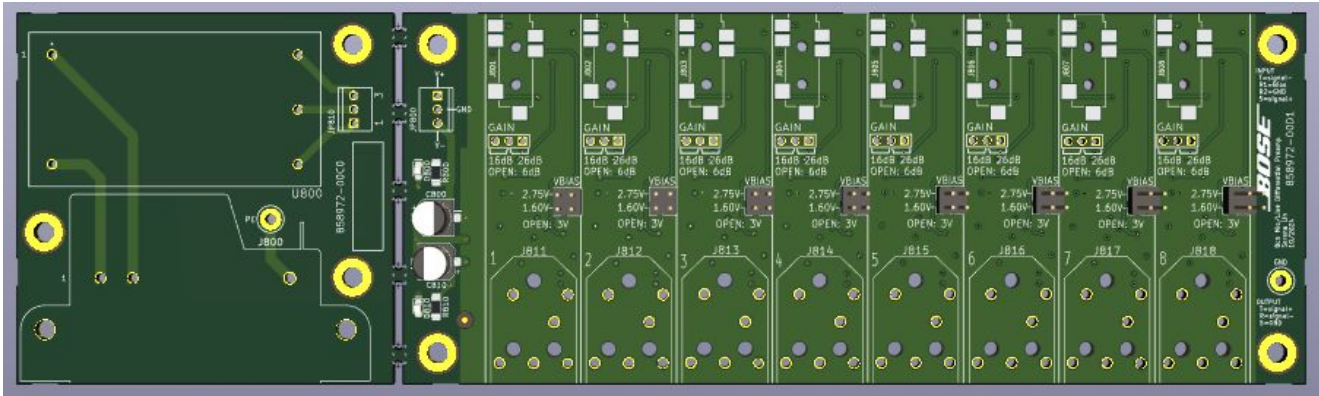
1 Channel Schematic Re-Design



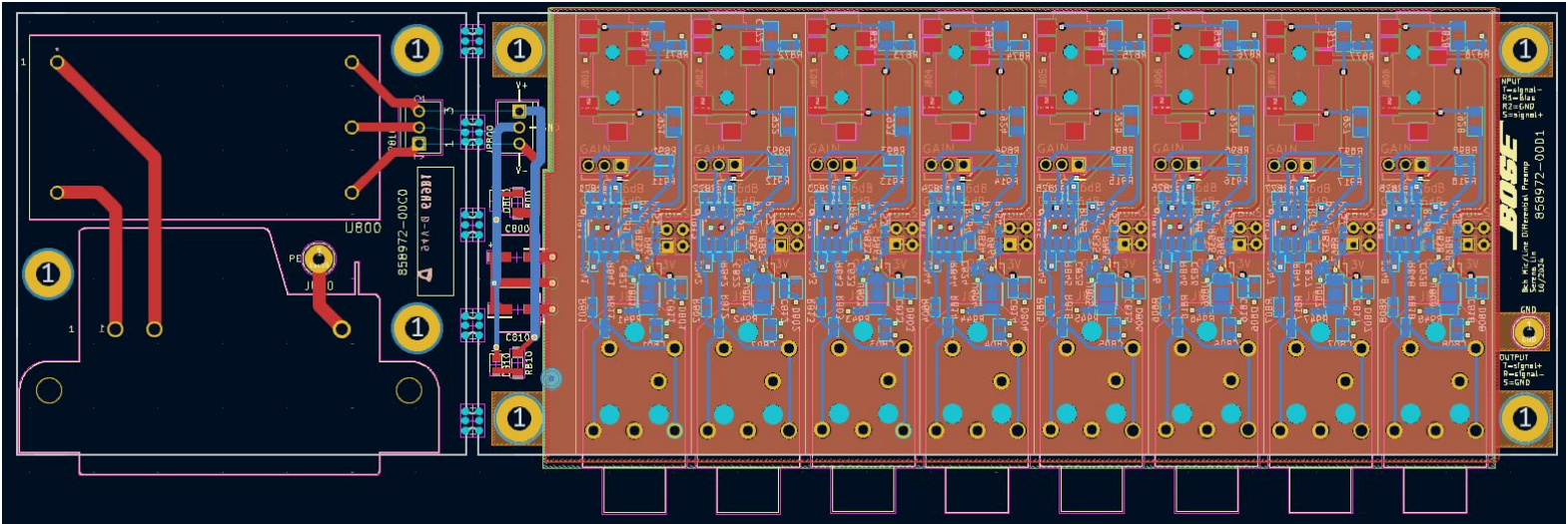
LAYOUT

Thank you Mika Ichiki-Welches!

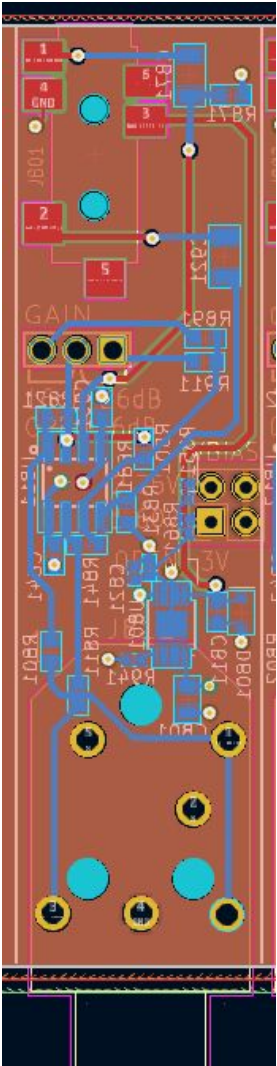
Layout: KiCad



Ki Cad 3D Viewer



Layout View 8 Channels & Power Supply

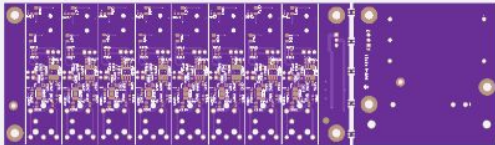
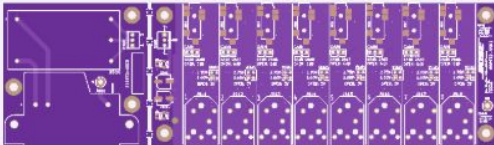


Single Channel View

ORDER PCB & COMPONENTS

JLCPCB

Top Designator	Bottom Designator	Comment	Footprint	Matched Part Detail	Qty	Source	Lib Type	Total Cost	Select
C800,C810		100uF	CP_Elec_8...	PM101M025E077PTR C2858382 25V 100uF ±20% SMD,8.3x7.7mm Polyme...	10	JLCPCB	Extended	\$1.8000	☒
	C811,C812,C813,...	4.7uF	0803	CL10A475KQ8NNNC C8032 6.3V 4.7uF X5R ±10% 0803 Multilayer Cer...	50	JLCPCB	Extended	\$0.2050	☒
	C821,C822,C823,...	0.47uF	0402	EMK105ABJ474KV-F C2169173 16V 470nF X5R ±10% 0402 Multilayer Cer...	50	JLCPCB	Extended	\$0.4100	☒
	C841,C842,C843,...	0.1uF	0803	C0603C104K5RACAUTO C129821 50V 100nF X7R ±10% 0803 Multilayer Cer...	90	JLCPCB	Extended	\$1.1250	☒
	C871,C872,C873,...	22uF	1206	C1206X8S226K8R3NT C2844315 6.3V 22uF ±10% 1206 Multilayer Ceramic ...	82	JLCPCB	Extended	\$4.1328	☒
D800,D810		GREEN	0805	SZYY0805YG C434435 Yellow Green 0805 LED Indication - Discre...	20	JLCPCB	Extended	\$0.2100	☒
	D801,D802,D803,...	DZener...	D_SOD-32...	B2T52C3V3S-7-F C177018 Independent Type 3.1V~3.5V 200mW 3.3V...	48	My Parts...	Extended	\$0.00	☒
J800		DD21.01...	DD21.0111...	DD21.0111.1111 C17207034 10A - DC Power Connectors ROHS	5	My Parts...	Extended	\$0.00	☒
J801,J802,J803,J...		~	SJ-43516-...	SJ-43516-SMT-TR-PI C4992481 -20°C~+85°C 12V Silver 1A - Audio Conne...	40	My Parts...	Extended	\$0.00	☒
J811,J812,J813,J...		~	SWITCHC...	RN112BPC C4992274 Silver Copper Alloy - Audio Connectors (H...	40	My Parts...	Extended	\$0.00	☒
J834,J831,J83...		Conn_02...	PinHeader...	ZX-P22.54-2-2PZZ C7501273 Gold 1KV 3A Direct Insert Policy 2.5mm 4P...	41	JLCPCB	Extended	\$1.4719	☒
JP800,JP810		640466-3	TE_64046...	640466-3 C86503 1x3P 3P 1.254mm 3 Brass Alloy PlugIn,P...	11	JLCPCB	Extended	\$1.1385	☒
JP821,JP822,JP8...		640466-3	AMPHEN...	68001-203HLF C6348216 3A Direct Insert 3P -65°C~+125°C 2.54mm ...	40	My Parts...	Extended	\$0.00	☒
R800,R810		15k	1206	RC1206FR-0715KL C114932 250mW Thick Film Resistors 200V ±100pp...	20	JLCPCB	Extended	\$0.0880	☒
	R801,R802,R803,...	100	0803	MR08X1000FTL C327825 125mW Thick Film Resistors 75V ±100pp...	90	JLCPCB	Extended	\$0.1800	☒
	R821,R822,R823,...	10k	0803	PTFR0803B10K0P9 C328421 100mW Thin Film Resistor 100V ±25ppm/...	82	JLCPCB	Extended	\$8.3222	☒
	R831,R832,R833,...	30k	0402	SY0402DE30KP C2763772 62.5mW Thick Film Resistors 50V ±0.5% ±...	50	JLCPCB	Extended	\$0.1450	☒
	R851,R852,R853,...	330k	0402	0803WAF3303T5E C23137 100mW Thick Film Resistors 75V ±100pp...	50	JLCPCB	Basic	\$0.0500	☒
	R861,R862,R863,...	34.8k	0402	0803WAF3482T5E C23141 100mW Thick Film Resistors 75V ±100pp...	50	JLCPCB	Extended	\$0.0500	☒
	R871,R872,R873,...	100k	0803	RC0803FR-07100KL C14675 100mW Thick Film Resistors 75V ±100pp...	90	JLCPCB	Extended	\$0.0810	☒
	R891,R892,R893,...	4.64k	0803	RS-03K4641FT C321920 100mW Thick Film Resistors 50V ±100pp...	50	JLCPCB	Extended	\$0.0250	☒
	R901,R902,R904,...	20k	0803	AC0803DR-0720KL C227463 100mW Thick Film Resistors 75V ±100pp...	45	JLCPCB	Extended	\$0.1845	☒
	R903	20k	0803	0803WAF2002T5E C4184 100mW Thick Film Resistors 75V ±100pp...	20	JLCPCB	Basic	\$0.0200	☒
	R911,R912,R913,...	1.1k	0803	0803WAF1101T5E C22784 100mW Thick Film Resistors 75V ±100pp...	50	JLCPCB	Extended	\$0.0500	☒
	R941,R942,R943,...	180	0402	WR04X181JTL C172158 62.5mW Thick Film Resistors 50V ±100pp...	50	JLCPCB	Extended	\$0.0850	☒
U800		~	RAC06-15...	RAC06-15DC C8638914 DIP-5 Power supply module,Power Suppli...	5	My Parts...	Extended	\$0.00	☒
	U801,U802,U803,...	LT3042E...	LT3042EM...	LT3042EMSE#TRPBF C461518 200mA Adjustable 0V~15V Positive electro...	40	JLCPCB	Extended	\$137.6400	☒
	U811,U812,U813,...	TL072	TL072IDT	TL072IDT C107648 3mV 4MHz 20pA SOIC-8 FET Input Amplifi...	40	JLCPCB	Extended	\$11.5800	☒
	C801,C802,C803,...	4.7uF	0805	FCC0805X475K250FT C5137571 25V 4.7uF X5R ±10% 0805 Multilayer Cer...	50	20 JLCPCE	Extended	\$0.2120	30 short



[Back to Upload File](#) Detected 4 layer board of 68.6x239.3mm(2.7x9.42 inches). [Gerber Viewer](#)

Base Material

FR-4

Flex

Aluminum

Copper Core

Rogers

PTFE Teflon

Layers

1

2

4

High Precision PCB

6

8

10

12

14

16

More

Dimensions

239.3

*

68.6

mm

PCB Qty

5

Product Type

Industrial/Consumer electronics

Aerospace

Medical

PCB Specifications

Different Design

1

2

3

4

Delivery Format

Single PCB

Panel by Customer

Panel by JLCPCB

PCB Thickness

0.4

0.6

0.8

1.0

1.2

1.6

2.0

PCB Color

Green

Purple

Red

Yellow

Blue

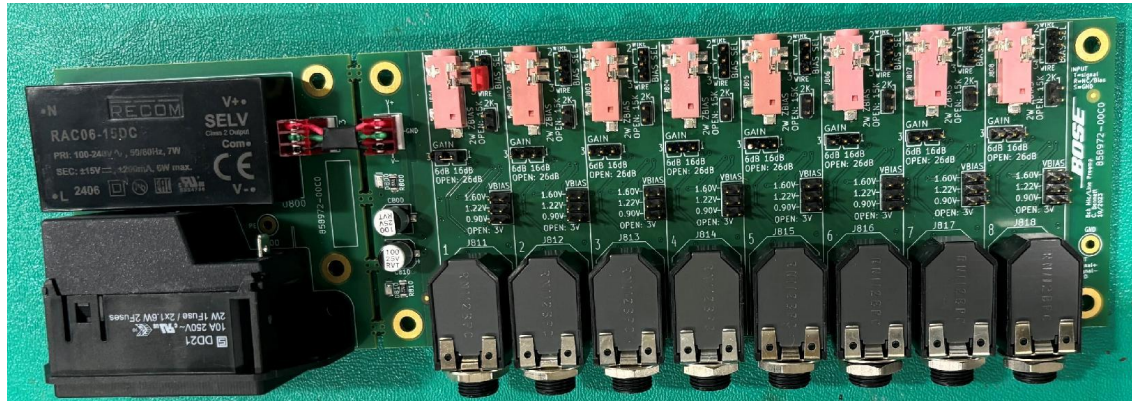
White

Black

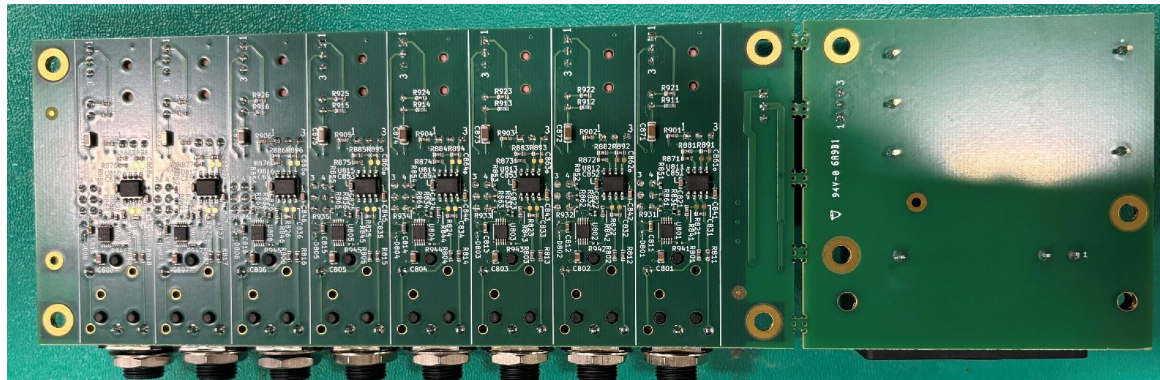
Silkscreen

White

Single Ended Preamp

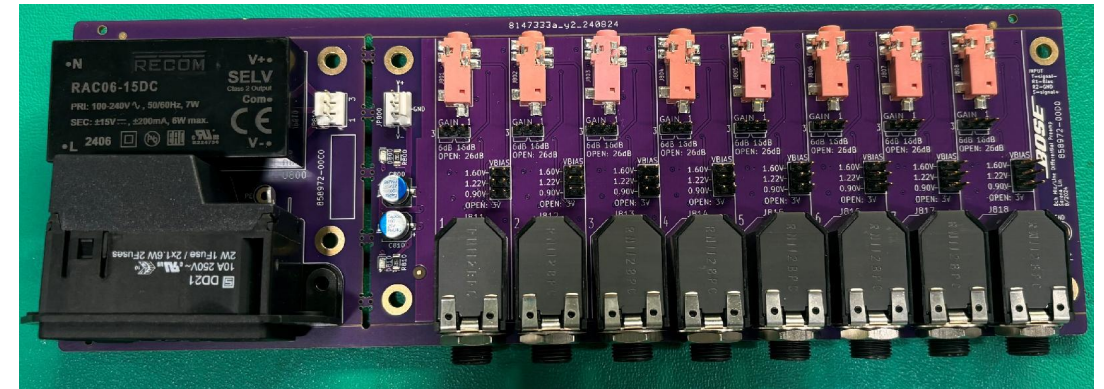


Single Ended Pre-Amp Front View

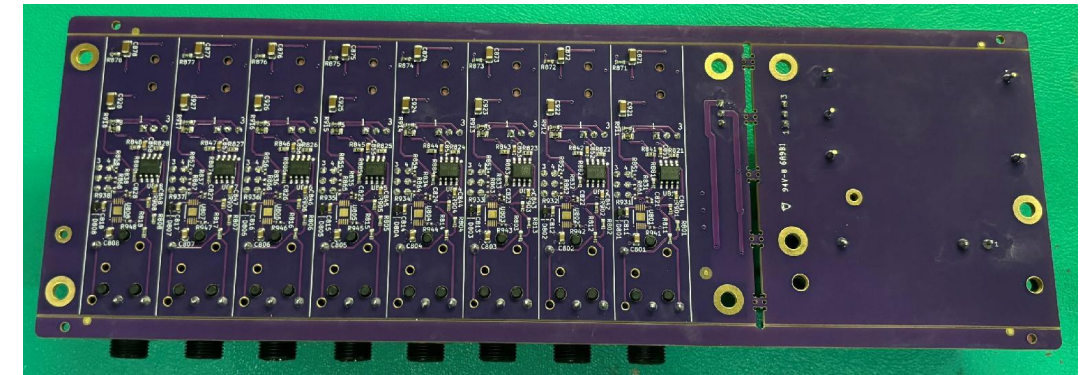


Single Ended Pre-Amp Back View

Differential Preamp



New Differential Pre-Amp Front View

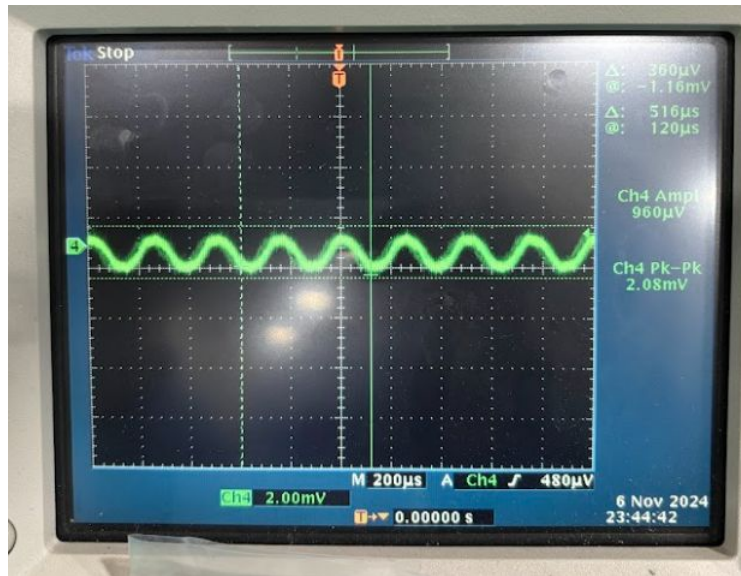


New Differential Pre-Amp Back View

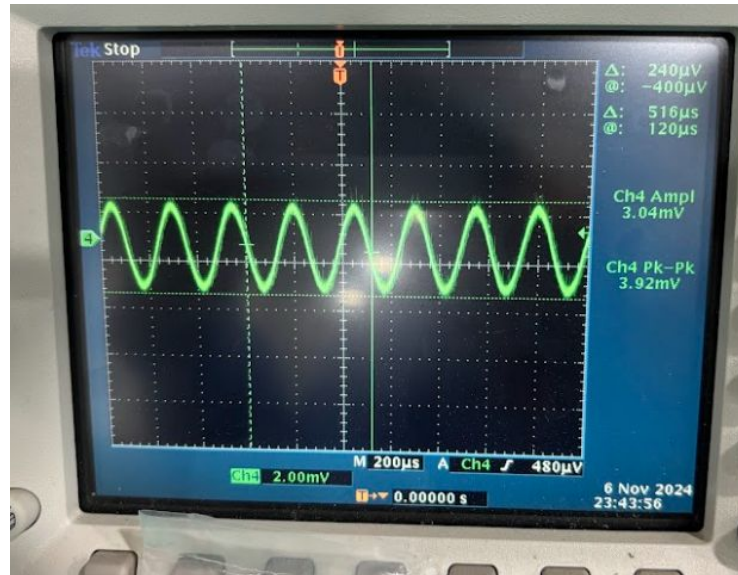
TESTING

PRE-AMPS

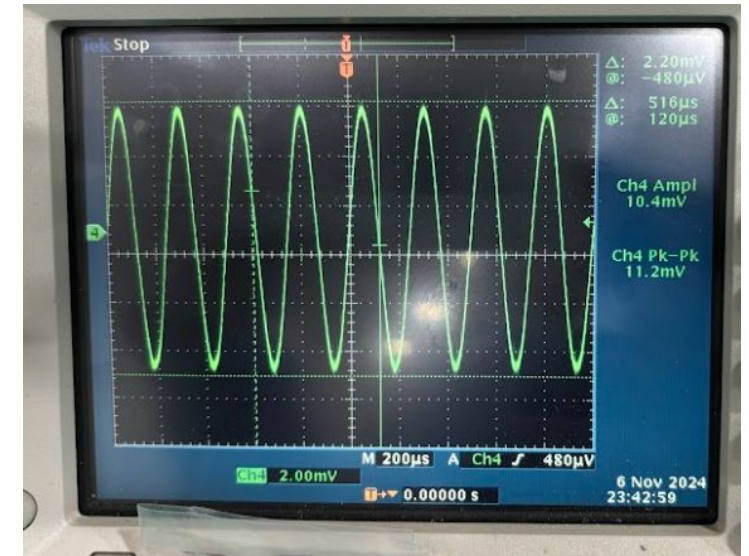
Differential Preamp Benchtop Testing: Infineon Mic



6 dB



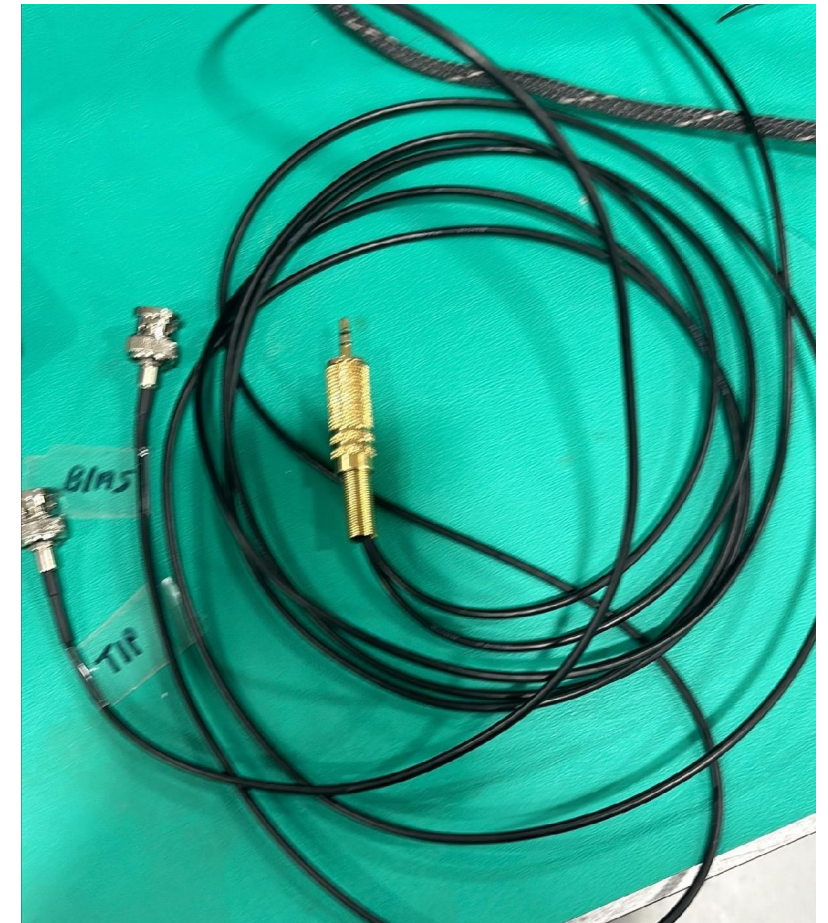
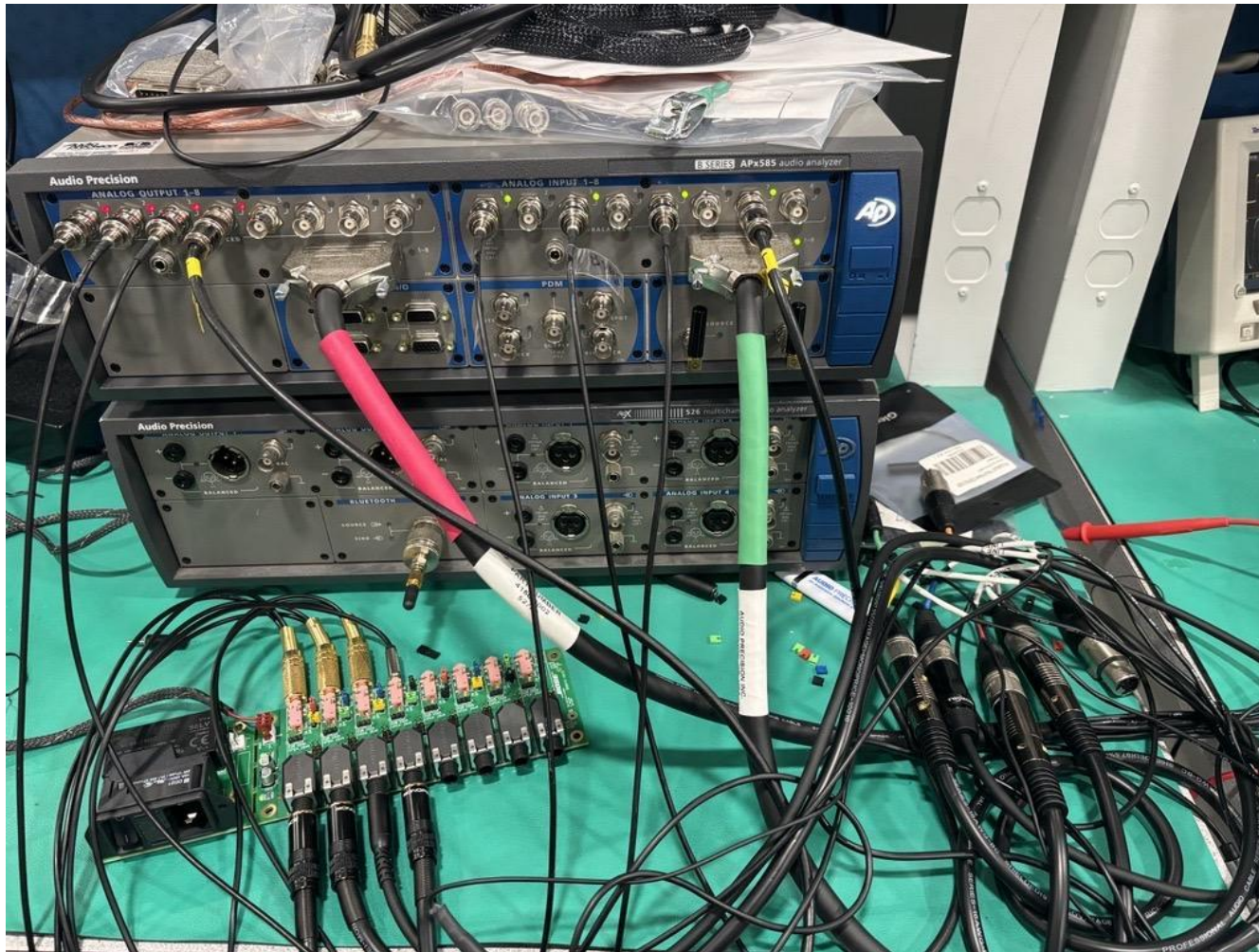
16 dB



26 dB

Thank you Matthew McKee!

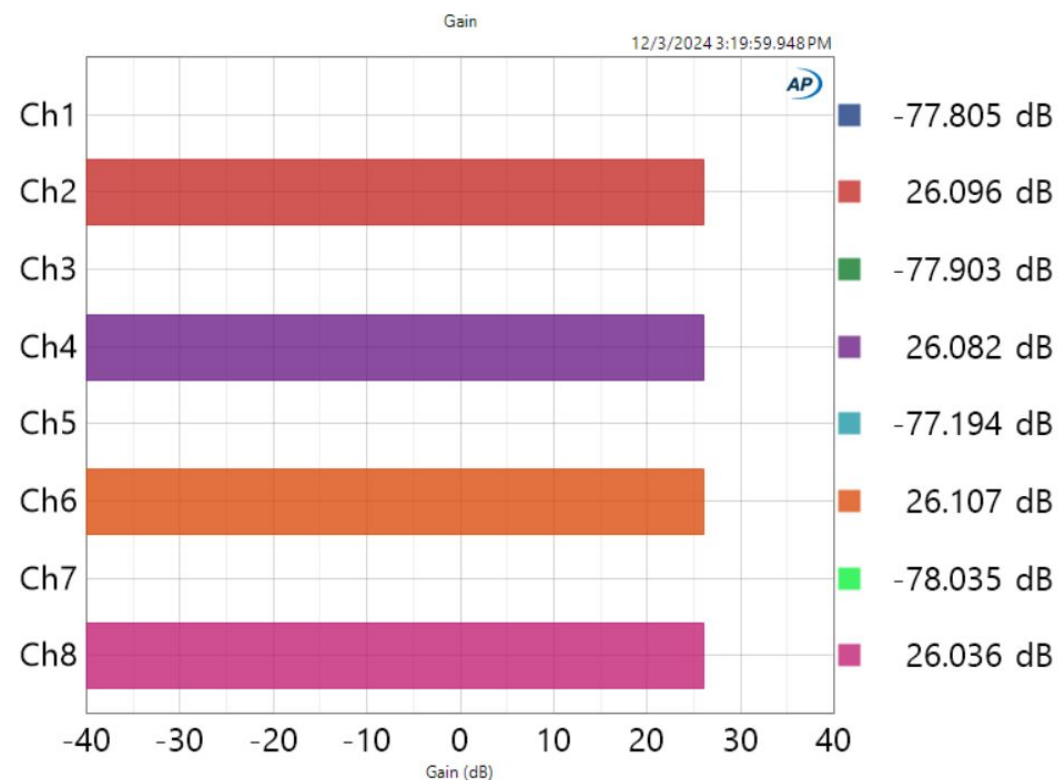
Testing ooCo with AP



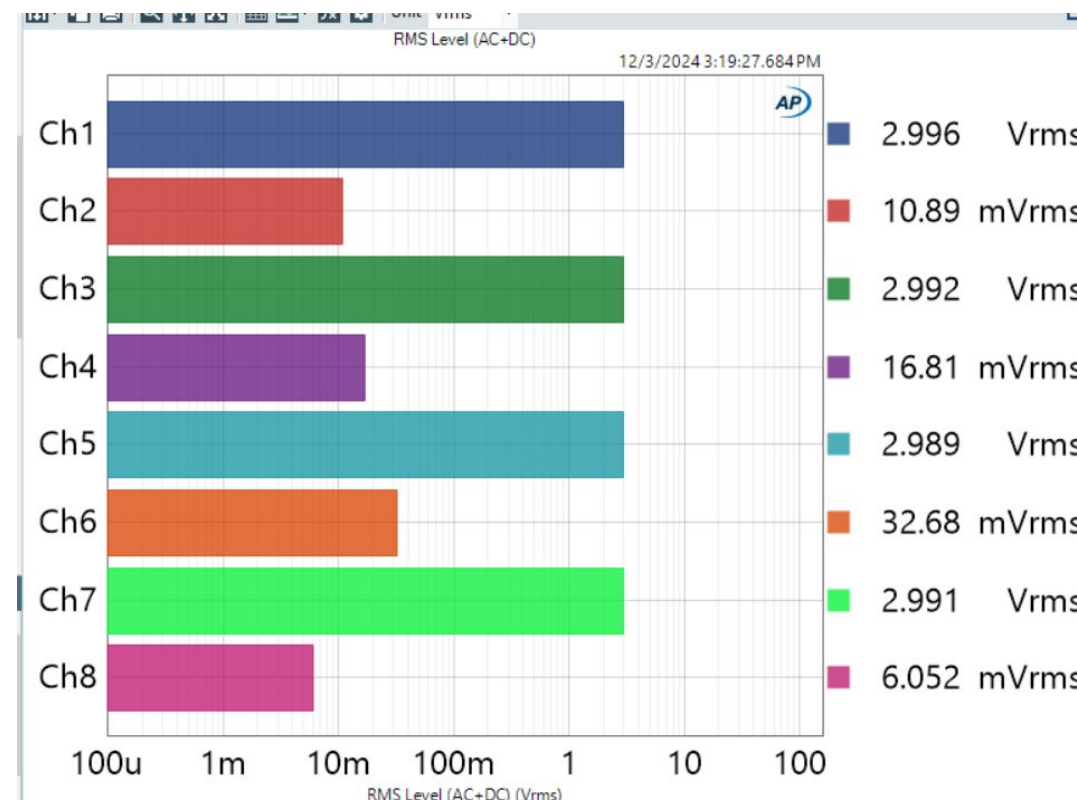
Custom TRS to 2 BNC Cable

Thank you Matt Irving!

Designing Test for ooCo with New AP



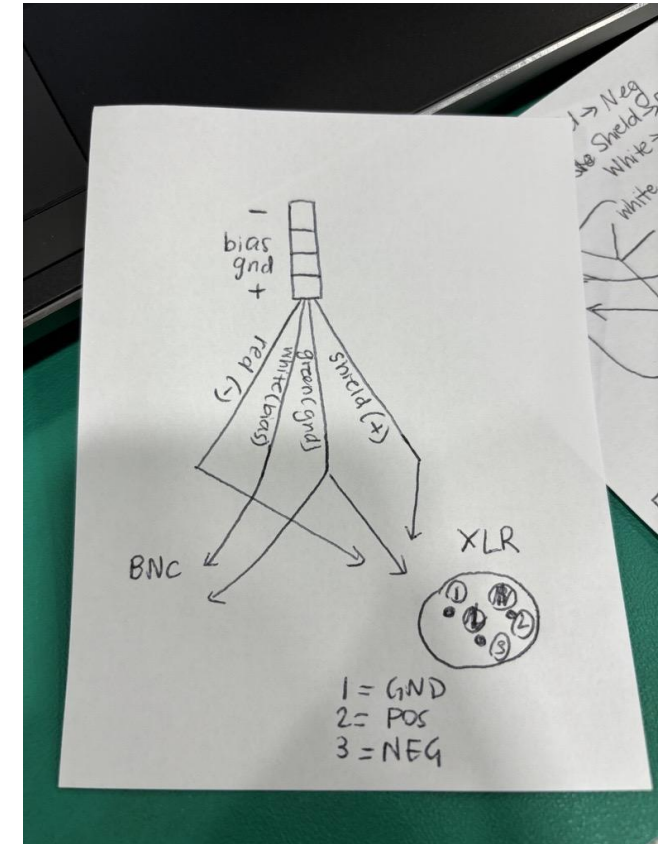
Testing Gain 26dB: 2 Channels



Testing Voltage Bias 3V: 2 Channels

Thank you Ken Bernier!

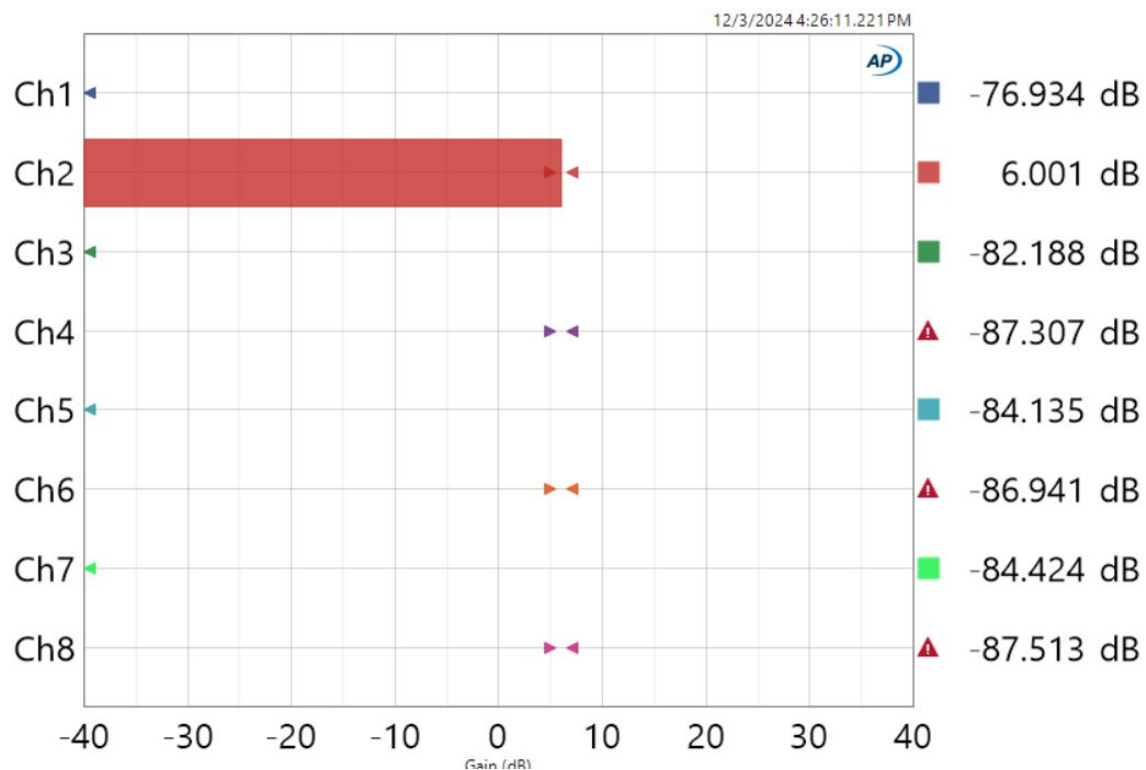
Testing rev D with AP



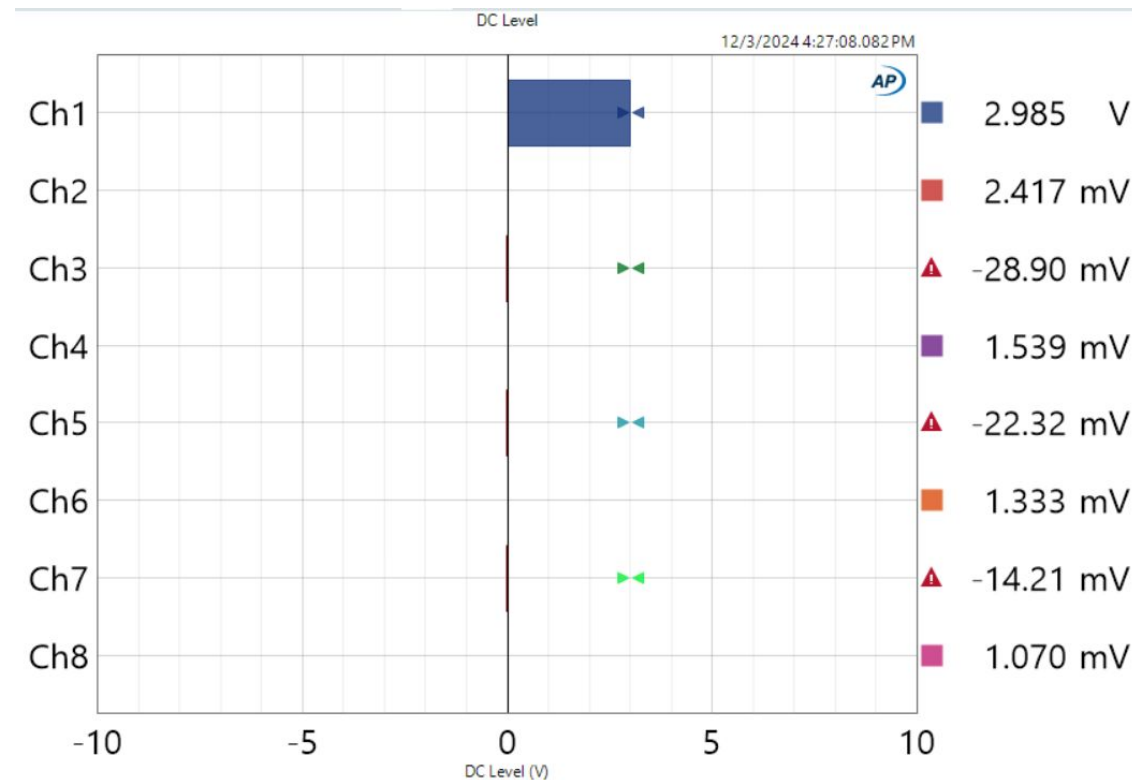
Schematic for Custom TRRS
1 BNC & 1XLR Cable

Thank you Andrew Munro!

Designing Test for 00D1 with 8 Ch AP



Testing Gain 6dB: 1 Channel



Testing Voltage Bias 3V: 1 Channel

Report with AP

Sequence Report

AP

Pre-Sequence Inputs:

directory name: AP testing

Summary

MicBias

Level and Gain	✓ PASSED
THD+N	✓ PASSED
Bias Voltage	✓ PASSED
Noise Spectra	✓ PASSED
Bias Noise	✓ PASSED

Sequence Result:

Sequence Result: ✓ PASSED

Thank you Oscar Mendoza
& Eric Mongeon!

Build



Next Steps!

- Enables R&D to use diff mics in products!
- Advertising to R&D
- Useful for people testing with analog mics
 - MDAQ, MOTU, RME, PREQ, and Audio Precision
- Successful MDAQ Testing with Richard Herr!

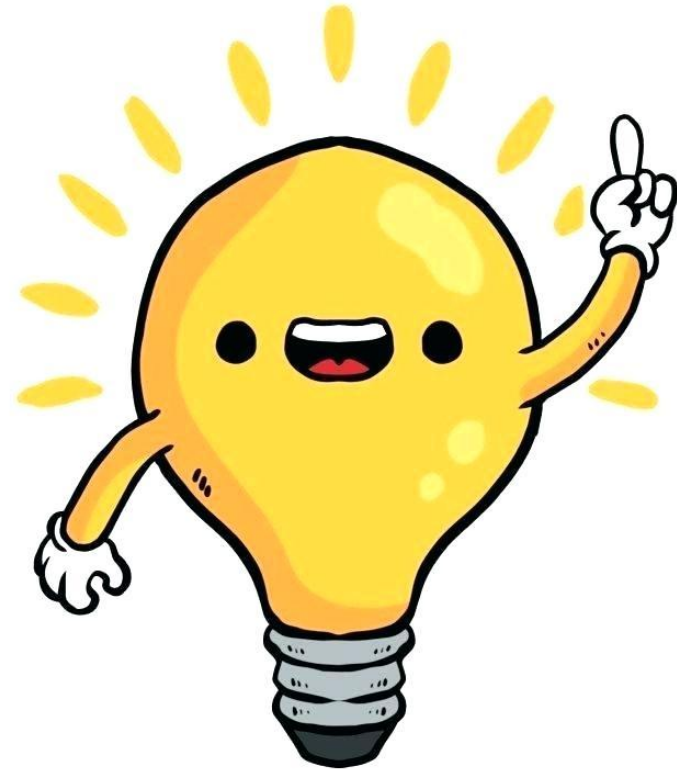


4

CONCLUDING THOUGHTS

Technical Learning

- KiCad & PCB Design
- PCB Assembly
- Soldering training
- Schematics
- Audio Precision
- Matlab & MDAQ
- Software M5 Arduino Bluetooth
- Raspberry Pis



Additional Learnings!

- Practical skills
 - Debugging!!
 - Systems Design
 - Asking Questions
- First co op!
- Coffee Chats!
- I love EE & CE!
- Research Interests

11/6 11:51 AM



Testing is going really well !

Ichiki-Welches, Mika 11/6 11:52 AM

Amazing news! We're back on track and learning a lot!



OVERALL --- STATISTICS

OFFICIAL STATS

2



Number of times
shocked on the
preamp



Percentage of times
the word "Serena" was
used to refer to me

27



Number of grey hairs
given to Mika

43%



Body Percentage of Free
Hot Chocolate Post Co-Op

5

Number of times
packages got stuck
in customs!



1

Number of times maxed out
Mika's Corporate Credit Card

 **Irving, Matthew**

Chat Shared +

🗨️ ⌵ 👤 ⋮ 📎

Tuesday, November 12

Irving, Matthew 11/12 2:05 PM

Yo do you understand circuits?

i do actually



👍 ❤️ 😂 🤔 🗨️ ⌵ ⋮

Love to see it! haha

😂 1

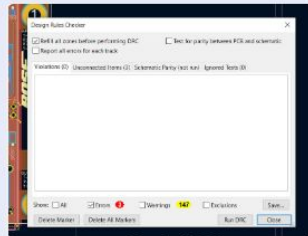


Ichiki-Welches, Mika

Chat Shared +

🗨️ ⌵ 👤 ⋮ 📎

10/2 4:22 PM



YAYYYYY

❤️ 1



Hudson, Cameron

Chat Shared +

🗨️ ⌵ 👤 ⋮ 📎



me when the op amp i select has 45x less noise than the one before

😂 1

I'm out of control w these memes

Hudson, Cameron 8/14 10:51 AM



HARDWARE NO NO STOP TALKING ABOUT

😂 1

I'M SO SCARED



Chambers, Thomas

Chat Shared +

🗨️ ⌵ 👤 ⋮ 📎

Chambers, Thomas 11/1 10:40 AM



did I go too far with colors lol?

👍 ❤️ 😂 🤔 🗨️ ⌵ ✎ ⋮

LOL I like it 🤔 but I am not to be trusted with matters regarding colors 😂

😂 1

Chambers, Thomas 11/1 10:56 AM

I dialed it back a bit

Hudson, Cameron 8/15 10:58 AM

Just went to shake the security guard Mike's hand and he hit me with a two piece dap I'm done girl

❤️ 1

Hudson, Cameron 8/15 11:23 AM

New song

8/15 2:49 PM

Hudson, Cameron 8/15/2024 10:58 AM
Just went to shake the security guard Mike's hand and he hit me with a two piece dap I'm done girl

Peak career moment

❤️ 1

udson, Cameron 8/6 10:52 AM

Crazy that they had a sticker just for this situation



OMG ITS WORKING

that is awesome thank you 😊

Thank you!

- Bruce McCauley
- **Mika Ichiki-Welches**
- Kemal Kulovic
- Thomas Chambers
- Cameron Hudson
- Matt Irving
- Eric Mongeon
- Oscar Mendoza
- Ken Bernier
- Matthew McKee
- Richard Herr
- Andrew Munro
- Jonathan Grovesteen
- Becky Gavin
- Lizzie Dings
- Richie Lanoue
- Zachary McHugh
- Caitlin Hanson
- David Lozupone
- Alvaro Alcocer
- Ben Hodgkins
- Roberto Gonzalez
- Lauren Westendorf
- Kyle Ritchie
- Cristian Carrasco
- Matt Smith
- Todd McFadden
- Mark McCauley
- Ric Carreras
- Frank Braney
- Douglas Morton
- Manuela Silva
- Pedro Teixeira
- Amin Ulhaq
- Rapid Proto Team!
- API Book Club!

Everyone else that has helped or guided me along the way!